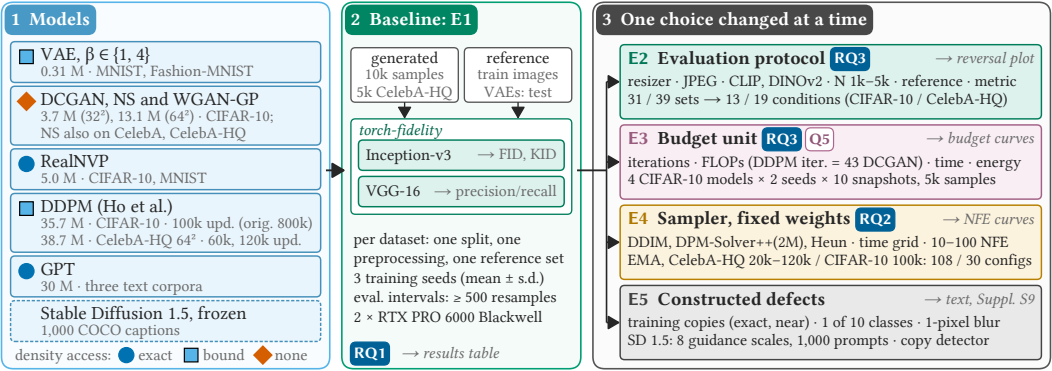
1 Models
VAE, β ∈ {1, 4}
0.31 M · MNIST, Fashion-MNIST
DCGAN, NS and WGAN-GP
3.7 M (32²), 13.1 M (64²) · CIFAR-10;
NS also on CelebA, CelebA-HQ
RealNVP
5.0 M · CIFAR-10, MNIST
DDPM (Ho et al.)
35.7 M · CIFAR-10 · 100k upd. (orig. 800k)
38.7 M · CelebA-HQ 64² · 60k, 120k upd.
GPT
30 M · three text corpora
Stable Diffusion 1.5, frozen
1,000 COCO captions
density access: exact bound none
2 Baseline: E1
generated
10k samples
5k CelebA-HQ
reference
train images
VAEs: test
torch-fidelity
Inception-v3 → FID, KID
VGG-16 → precision/recall
per dataset: one split, one
preprocessing, one reference set
3 training seeds (mean ± s.d.)
eval. intervals: ≥ 500 resamples
2 × RTX PRO 6000 Blackwell
RQ1 → results table
3 One choice changed at a time
E2 Evaluation protocol RQ3 → reversal plot
resizer · JPEG · CLIP, DINOv2 · N 1k–5k · reference · metric
31 / 39 sets → 13 / 19 conditions (CIFAR-10 / CelebA-HQ)
E3 Budget unit RQ3 Q5 → budget curves
iterations · FLOPs (DDPM iter. = 43 DCGAN) · time · energy
4 CIFAR-10 models × 2 seeds × 10 snapshots, 5k samples
E4 Sampler, fixed weights RQ2 → NFE curves
DDIM, DPM-Solver++(2M), Heun · time grid · 10–100 NFE
EMA, CelebA-HQ 20k–120k / CIFAR-10 100k: 108 / 30 configs
E5 Constructed defects → text, Suppl. S9
training copies (exact, near) · 1 of 10 classes · 1-pixel blur
SD 1.5: 8 guidance scales, 1,000 prompts · copy detector